this test was the next best thing to being able to compare string theory with experiment. Moreover, Candelas's results went far beyond the single numerical result that Ellingsrud and Strømme claimed to have calculated. He and his collaborators claimed to have also answered many other questions that were tremendously more difficult—so difficult in fact, that no mathematician had ever even attempted to address them. But could the string theory results be trusted? The meeting ended with a great deal of fruitful exchange between mathematicians and physicists, but no resolution of the discrepancy.

About a month later, an e-mail message was widely circulated among participants in the Berkeley meeting with the subject heading Physics Wins! Ellingsrud and Strømme had found an error in their computer code that, when corrected, confirmed Candelas's result. Since then, there have been many mathematical checks on the quantitative reliability of the mirror symmetry of string theory: It has passed all with flying colors. Even more recently, almost a decade after physicists discovered mirror symmetry, mathematicians have made great progress in revealing its inherent mathematical foundations. By utilizing substantial contributions of the mathematicians Maxim Kontsevich, Yuri Manin, Gang Tian, Jun Li, and Alexander Givental, Yau and his collaborators Bong Lian and Kefeng Liu have finally found a rigorous mathematical proof of the formulas used to count spheres inside Calabi-Yau spaces, thereby solving problems that have puzzled mathematicians for hundreds of years.

Beyond the particulars of this success, what these developments really highlight is the role that physics has begun to play in modern mathematics. For quite some time, physicists have "mined" mathematical archives in search of tools for constructing and analyzing models of the physical world. Now, through the discovery of string theory, physics is beginning to repay the debt and to provide mathematicians with powerful new approaches to their unsolved problems. String theory not only provides a unifying framework for physics, but it may well forge an equally deep union with mathematics as well.